

Online, In-Person, or Both?

How Communication Channels Relate to Undergraduate Friendship Quality

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Introduction

Our survey examined the strength of close friendships maintained primarily online, primarily in person, and through a combination of both. Friendship strength was measured using self-reported friendship quality (a five-item composite) and communicative competence (a four-item composite adapted from the [Communicative Competence Scale](#) [Wiemann, 1977; Rubin, n.d.]). We also assessed the frequency of communication across each channel. Prior research has shown that social networking sites and instant messaging platforms can facilitate communication maintenance through “presence awareness.” However, these studies have mainly focused on strengthening existing friendships rather than forming new ones ([Yu & Leung, 2023](#)). Our data also include information on platform use and demographic variables to allow subgroup analysis, such as examining whether men report more gaming-based friendships. The findings aim to provide a clearer understanding of how communication channels and digital platforms support or hinder the quality of friendships among undergraduates.

Research Questions and Hypotheses

Research Question 1: Do close friendships maintained mostly online, mostly in-person, or a mix of both differ in overall friendship quality?

- Due to the varying levels of intimacy in these differently maintained friendships, the effectiveness of each in dealing with relationship difficulties may differ based on the foundation of the friendship.

Hypothesis 1: Students who primarily communicate in person with their closest friends will report higher quality friendships than those who primarily communicate online.

- **Independent Variable:** Primary communication channel
- **Dependent Variable:** Friendship quality composite

- **Rationale:** In-person communication allows for deeper understanding within friendships through actions such as emotional expressiveness, nonverbal cues, and shared physical and emotional experiences. These factors often foster a stronger sense of trust and connection in relationships compared to those maintained primarily online.

Research Question 2: How do students' feelings of connectedness and perceived effectiveness differ when they communicate online versus in person with their closest friend group?

- This question focuses on whether students feel differently connected to their closest friends and view their communication as more or less effective depending on the locational context of those interactions.

Hypothesis 2: People will feel more connected to their closest friend group and perceive their communication as more effective when they interact in person as opposed to online.

- **Independent Variable:** Communication mode (online vs in-person)
- **Dependent Variable:** Perceived connectedness & effectiveness ratings for each mode
- **Rationale:** Being able to communicate with friends in person provides more opportunities for accurate understanding through the use of facial expressions, tone, and body language. These nonverbal cues help clarify meaning and emotions, leading to stronger feelings of connection and more effective communication compared to online interactions.

Research Question 3: Is communicative competence associated with friendship quality in students' closest friend group?

- Finding differences in friendship quality based on communicative competence can help clarify how communication skills relate to interpersonal outcomes.

Hypothesis 3: Higher communicative competence will be positively related to higher friendship quality.

- **Independent Variable:** Communicative competence composite
- **Dependent Variable:** Friendship quality composite

- **Rationale:** Individuals who adapt and express themselves effectively tend to feel more comfortable with those around them and experience better interpersonal outcomes, including closer and more satisfying friendships.

Research Question 4: Does the use of gaming chats with friends differ by gender?

- Gaming platforms are typically male-dominated; we aim to analyze whether this is also evident among undergraduate students.

Hypothesis 4: Men will be more likely than women to report using gaming chats to communicate with their closest friends.

- **Independent Variable:** Gender
- **Dependent Variable:** Gaming Used
- **Rationale:** Gaming platforms and Discord-style chatting services are often male-dominated, so we expect a higher rate of gaming-based communication among male undergraduates.

Methods

Data Collection

Data was collected via an online survey using QuestionPro. We used convenience sampling to recruit undergraduate students at Virginia Tech through personal networks, social media, and group chats. Students received the link and were required to answer initial screening questions before participating in the study. Participation was voluntary and anonymous; no names or other identifying information were collected or distributed. Responses were stored in QuestionPro and exported for statistical analysis.

Participants

Our sample consisted of $N = 31$ undergraduate students at Virginia Tech. Participants were aged 18 to 23 years (Mean = 20.3, Standard Deviation = 1.16). Regarding gender, 17 identified as women (55%) while 14 identified as men (45%).

Most participants reported that their closest-friends group was located at their school ($n = 23$, 74%), followed by their hometown ($n = 4$, 13%) and another city or country ($n = 4$, 13%). On average, the survey took roughly 4 minutes to complete, as measured by QuestionPro's completion time.

Survey Construction

Our survey had five main sections. The first section assessed the primary communication channel and overall quality of communication with students' closest friends. This section aimed to determine whether people mainly communicate online or in person, and the quality of that communication. The second section was friendship quality using a five-item scale that assessed trust, understanding, acceptance, and willingness to be open. This section was to gauge the quality level of each person's friendships. The third was an online experience. In this section, we assessed the extent to which people communicate online, the platforms they use, and how effective that communication is. The fourth section focused on in-person communication: how often participants communicated in person and how this made them feel more or less connected, and the level of effectiveness of their communication. Our final section measured communication competence with four items adapted from the Communicative Competence Scale (Wiemann, 1977; Rubin, n.d.). We asked them questions about their friend group and had them answer each question on a 5-point scale from strongly disagree to strongly agree. Demographic items finished off the survey, asking about age, gender, year in school, and whether their primary friend group was located.

These sections allowed us to use the variables in our research questions and hypotheses. For our first research question, we asked if there's a difference in friendship quality depending on whether the friendship is strictly online, strictly in-person, or a mix of both. For this, we set the independent variable as the primary way of communication, and we set the dependent variable as the quality of the friendship. Our second research question asked how online and in-person communication relate to feelings of connectedness and perceived effectiveness. Here, the independent variable was communication mode, and the dependent variables were connectedness and effectiveness for each mode. Our third research question focused on whether communicative competence is associated with friendship quality; the independent variable was the communicative competence composite, and the dependent variable was the friendship quality composite. Our last question is whether the use of gaming chats differs by gender. The independent variable in this is the person's gender, and the dependent variable is whether gaming chats are used or not. We measured all of these by gathering and tallying the results of our survey.

Data Analysis

RQ1/H1: Primary channel and friendship quality (ANOVA)

We tested whether friendship quality differed based on primary communication channel (mostly online, mostly in-person, or mixed). A one-way ANOVA test was conducted with the primary channel as the independent variable and friendship quality (mean of five friendship quality items) as the dependent variable. Mean friendship quality was very similar across the three groups: mostly online ($M = 4.61$, $SD = 0.5$, $n = 7$), mostly in-person ($M = 4.43$, $SD = 0.67$, $n = 6$), and mixed ($M = 4.51$, $SD = 0.45$, $n = 17$). The ANOVA showed no significance ($p = 0.82$), and Tukey tests revealed no significant pairwise differences (all $p > 0.8$). Thus, H1 was not supported; in this sample, students reported similarly high friendship quality regardless of whether they primarily communicated with their closest friend group online, in-person, or in a mixed capacity.

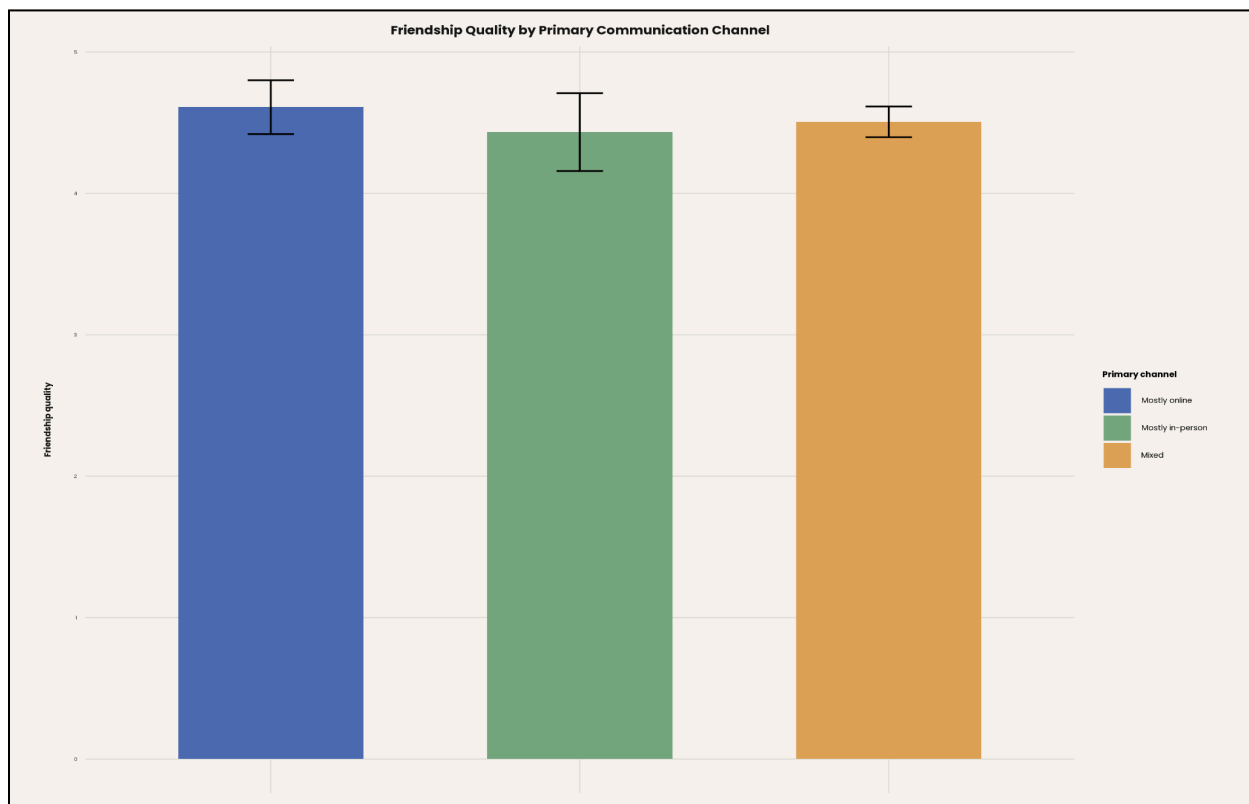


Figure 1: Friendship quality by primary communication channel ($N = 30$).

RQ2/H2: Online vs in-person connectedness and effectiveness (Paired-samples t-test)

We compared students' ratings of connectedness and perceived effectiveness when communicating online versus in-person with their closest friend group using paired t-tests. Students felt far more connected when communicating in person ($M = 4.48$, $SD = 0.63$) than when communicating online ($M = 3.45$, $SD = 0.68$). This difference was statistically significant ($p < 0.001$) with a considerable within-person effect size ($d_x = 1.37$). Practically, an effect this large means that, for most students, in-person interactions with their closest friends felt noticeably more connecting than online conversations, not just slightly better.

Moreover, students also rated their in-person communication as more effective ($M = 4.29$, $SD = 0.78$) than their online communication ($M = 3.55$, $SD = 0.77$). This difference was significant ($p < 0.001$) with a large effect size ($d_x = 0.87$). Combined, these results strongly support H2; we can reject the null and conclude that participants experienced communication with their closest friend group as more connecting and effective in-person than online.

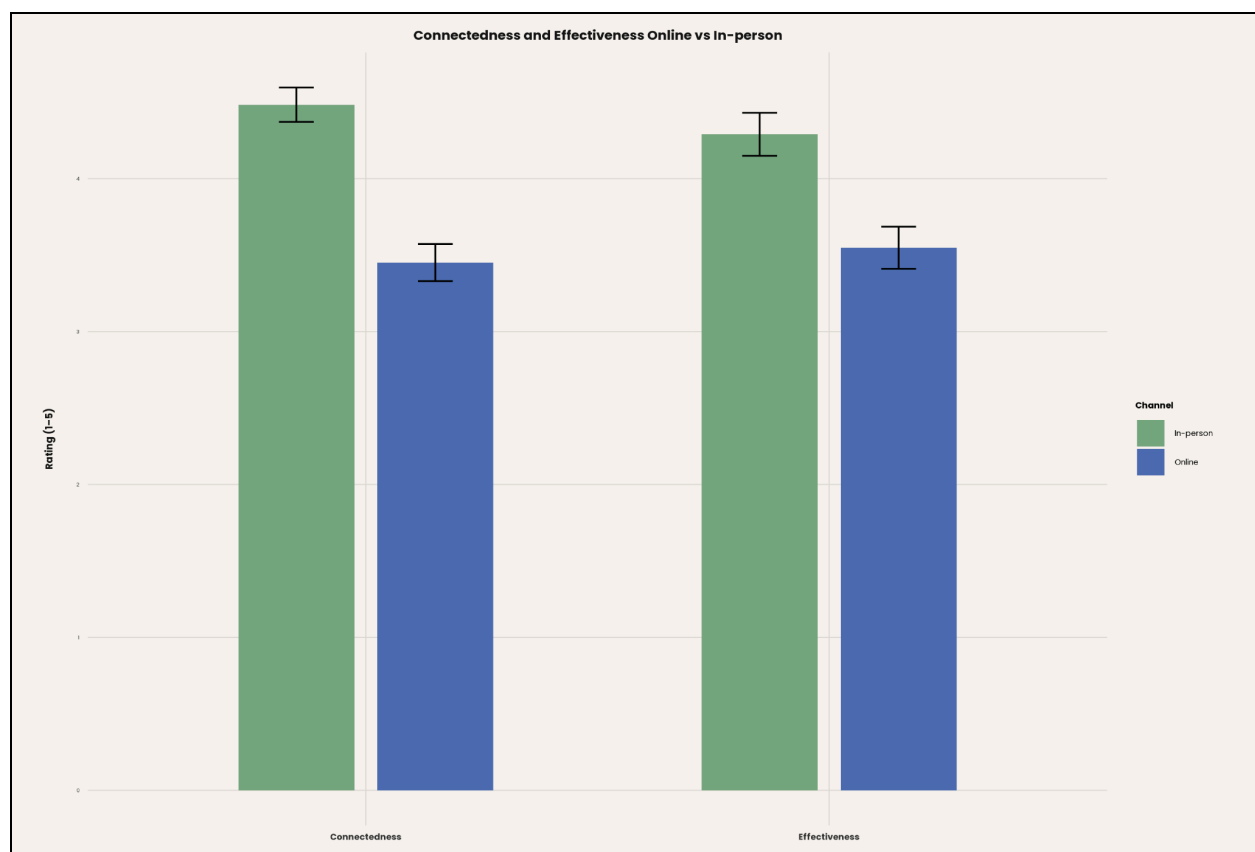


Figure 2: Connectedness & effectiveness online vs in-person ($N = 31$).

[RQ3/H3: Communicative competence and friendship quality \(Pearson correlation\)](#)

We examined whether communicative competence was associated with friendship quality. Communicative competence was measured with a four item composite (the mean of “I express my ideas clearly to this group,” “I adapt how I communicate to fit this group and the situation,” “I pick up on this group’s cues and respond appropriately,” and “I feel relaxed and comfortable when talking with this group,” adapted from a [verified communicative competence scale](#) [Rubin, n.d.]), and friendship quality. A Pearson correlation test revealed a small, non-significant positive relationship between competence and friendship quality, $p = 0.35$, and a 95% confidence interval of $[-0.19, 0.5]$. As a result, H3 was not supported as we failed to reject the null hypothesis. In this sample, students who saw themselves as more communicatively competent did not reliably report higher friendship quality with their closest friend group.

[RQ4/H4: Gender and gaming-based communication \(Chi-Square & Fisher’s Test\)](#)

We investigated whether using gaming chats with friend groups differed by gender. A dichotomous variable was created to track whether participants selected “video game chats” as a primary platform for communication (yes = 1, no = 0). Among women, 1 of 17 (5.9%) reported using gaming chats, compared to 5 of 14 men (35.7%). A chi-square analysis found a significant association, $p < 0.05$. However, the chi-square test assumes that expected counts in each cell are larger than 5. In our table, at least one cell failed this assumption, which means the p-value may not be accurate. Therefore, we rely on Fisher’s exact test, which indicated that the association did not reach conventional significance ($p = 0.067$). H4 was not supported; although men descriptively reported more gaming-based communication than women, our sample was too small to conclude that the difference is reliable.

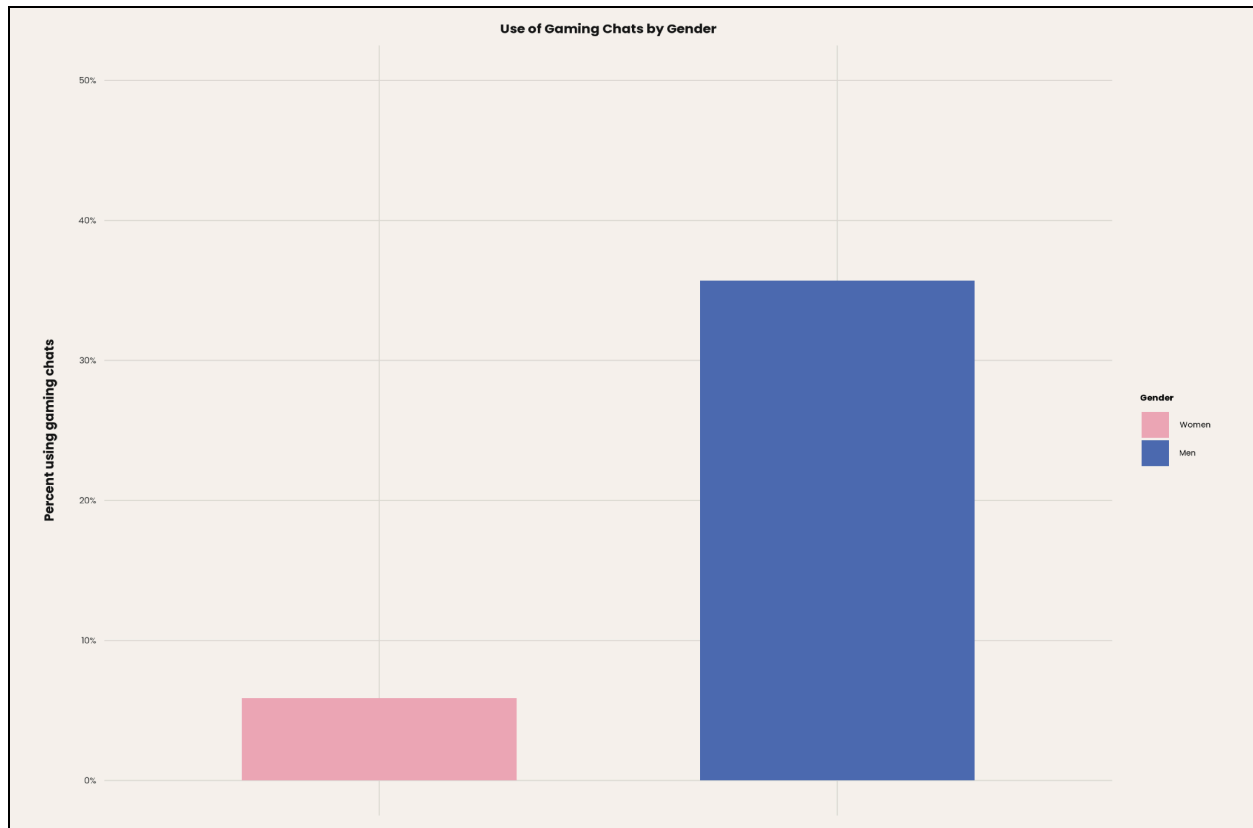


Figure 3: Gaming chat use across genders in our sample ($N = 31$).

Limitations

The main limitation that we faced was that our respondents had to meet specific criteria, such as being a college student, which narrowed the pool of individuals we could send the survey to. If we could conduct the survey differently, we would likely make it targeted to a broader audience so that we receive more responses. We could have opened the survey up to post-graduate students or grad students, which would still be targeted towards the college social experience. We would also choose a similar topic, but one with more clarity and conciseness. It also became hard to have respondents respond on time to meet our quota. In retrospect, we could have organized who we were specifically going to contact better and at an earlier date before creating the survey to meet our respondent goal. We could also have done more research on and familiarized ourselves with QuestionPro before creating the survey, making the process of collecting data and creating the survey through QuestionPro more seamless.

Conclusion

Through the delivery and analysis of this survey, we studied the relationship between different channels of communication and the strength of undergraduate friendships. We focused on understanding perceived closeness, communicative competence, and satisfaction gained within those friendships. Our findings showed that H1, H3, and H4 were not supported; however, H2 was. This information suggests that the communication channel does not significantly impact the quality of friendships in our sample. Despite this, it showed that respondents felt they had more effective and richer communication in person compared to online communication. Respondents who felt they were communicatively competent did not feel this had a significant impact on their friendships. The sample size was too small to determine whether the difference between men and women in gaming-chat use is statistically reliable. Overall, we found that undergraduate students maintain their friendships regardless of the form of communication they use. Online communication is effective in maintaining friendships; however, they can have deeper connections in person.

Team Member Contributions

- **Caleb:** Formulated research questions & hypotheses, developed 12 survey questions, rough outline for summary, brainstormed variables, revised summary, added references for communication competence scale, revised research questions & variables using Dr. Woods' recommendations, created questions using communication competence scale, created survey in QuestionPro, formulated data collection summary, performed data analysis & created visualizations, and revised final draft of the final report.
- **Carter:** Formulated research questions & hypotheses, developed 12 survey questions, rough outline for summary, brainstormed variables, revised survey questions, and conclusion for survey outline & final report.
- **Eve:** Formulated research questions & hypotheses, developed 12 survey questions, rough outline for summary, brainstormed variables, revised variables, completed limitations section.
- **Preston:** Formulated research questions & hypotheses, developed 12 survey questions, rough outline for summary, brainstormed variables, revised survey questions, and completed survey construction section.

- **Sydney:** Formulated research questions & hypotheses, developed 12 survey questions, rough outline for summary, brainstormed variables, revised research questions/hypotheses, crafted introduction, and completed participants section.

References

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AI Disclosure

We used OpenAI's ChatGPT (GPT-4) as a writing and question development aid. Specifically, we used it to:

- Help refine our research questions & hypotheses to clearly match the goals of our project
- Suggest small wording changes for survey items and scaling

The group made all final decisions regarding the research questions and survey wording.

Appendices

Appendix A: Survey Instrument

Appendix B: Survey Data Analysis

Appendix C: Survey Project Final Data